

Designing a Toolkit for Evaluating HCI Guidance Systems in Physical Spaces: Case of Flavoria Research Lunch Line

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Abstract

We report on the use case of an instrumented self-service lunch line of a research restaurant that gathers data about dining and diners, using scales, screens, and other collection instruments. The data collection is automated, but requires some participation or compliance from the diners. We developed and tested a non-intrusive LED cue system to guide diner behavior, evaluating its effectiveness through telemetry data, user observations and focus groups. Contrary to expectations, the system did not increase user compliance. To better understand user behavior, we applied context analysis, cognitive load, and affordance theories. The reflections of this research delivers a toolkit to support researchers in the human-computer interaction (HCI) field to analyze the factors for success/failure of guidance systems and/or guide the design of new such systems with high probability of success.

CCS Concepts

• Human-centered computing → Field studies.

Keywords

HCI, self service restaurants, toolkit, affordance, context, cognitive load, field study, LED

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1 Introduction

Designing effective human-computer interaction (HCI) in complex settings requires forming a comprehensive understanding of users' behavior and motivations, and taking into consideration the physical space and general context where the interaction takes place. The setting for our case study is this kind of a complex setting: Flavoria¹, a self-service lunch restaurant equipped with advanced instrumentation that can measure multiple aspects of the restaurant users' behavior, and screens that guide and inform the users in their interactions with the lunch line.

The goal of the instrumentation is to obtain data about trends and patterns in demand and nutrition, and about diners' behavior, so that the operation of self-service restaurants could be improved, for the diners, operators, and the environment. The broader context of our investigations on the HCI aspects of lunch lines is a large multidisciplinary research project *DiningFlow*² that works towards more sustainable self-service restaurants that better guide diners to make nutritionally advantageous choices.

Of the elaborate system of sensors and displays, central for the case study in this paper are the scales that weigh diners' plates before and after different food dispensing stations. Their purpose is to provide data from which it is, or should be, possible to determine

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¹<https://www.flavoria.fi/en/restaurant/>

²sites.utu.fi/diningflow/

the different kinds of foods on the diners' plates and the amounts of those foods. The main HCI aspect, or challenge, that this paper focuses on is guiding users to use the scales correctly so that the measurements are accurate.

The contributions of this case study to the HCI community are two-fold. First, we describe the design and implementation of a non-intrusive LED visual cue system to guide the diners' behavior with respect to tray placement. This field study provides general lessons learned regarding visual cue systems. Through evaluations via the system's telemetry data analysis and user observations, we hoped to confirm that user compliance to the data collection process would improve—but were unsuccessful. From this failure arises the second, more important contribution.

We describe a practical example of designing HCI systems and conducting HCI experiments in a very complex environment, in a real-life setting where “everything impacts everything” — and where the normal operation (of the restaurant) cannot be disturbed. In the face of such complexity, even seemingly isolated tasks, such as developing an effective cue system, cannot be tackled like that, in isolation. It became clear that to understand the diners' behavior well enough for designing efficient HCI, we needed to deploy a multidisciplinary research team and apply several supporting theories, including cognitive load theory and cognitive mapping, affordance theory, and theories of physical wayfinding. In doing so, we distilled this multifaceted approach to a framework, or toolkit, that can guide others facing similar challenges. We believe the all-embracing approach we applied to the specific HCI challenge generalizes beyond lunch lines and even visual cue systems: the lessons learned from this case study can apply, for example, in systems where users need to be guided to behave in certain ways, where users are barely aware that they are interacting with a system, or where a trouble-free people flow is of importance.

2 Theoretical Foundations

2.1 Context

In the study of human behavior, context plays a critical role in shaping experiences, influencing decision-making, and moderating interactions. It can be defined as the set of circumstances or factors that surround and influence a particular event, behavior, or interaction [8]. This definition is comprehensive, encapsulating the physical environment, social setting, cultural background, and temporal conditions in which something occurs. Context shapes how people perceive, understand, and respond to situations, making it essential for interpreting meaning and behavior. To operationalize the concept of context, we can analyze it through three key dimensions: environmental and physical context, socio-cultural context and psychological context.

Environmental and physical context. Environmental context refers to the overall surroundings or conditions in which an interaction or behavior takes place. When discussing environmental context, the term encompasses diverse aspects such as weather, social dynamics, and cultural norms [5].

When the emphasis is on the physical context, our focus shifts specifically to the tangible and material aspects of the surroundings [5]. This includes the physical layout, objects, structures, spatial arrangements, and the physical properties of objects (e.g., size, shape, texture). For instance, the physical layout of a restaurant, including tables, chairs, the lunch line, and dishes (excluding the diners), shapes movement, interaction, and overall experience.

Socio-cultural context. Socio-cultural context refers to the social structures, cultural norms, beliefs, values, and practices that shape how individuals interpret their experiences and interact with the world around them [1]. This dimension of context includes factors such as language, traditions, social norms, power dynamics, religious beliefs, and community practices. It encompasses the influence of a person's social relationships and the cultural meanings assigned to objects, events, and behaviors.

Psychological context. Psychological context refers to the internal, mental, and emotional factors that influence how an individual perceives, interprets, and responds to situations, events, or interactions [2]. It encompasses a person's thoughts, emotions, beliefs, motivations, and past experiences. The role of emotions and evaluation of them in various contexts is widely recognised [3, 6, 19, 20, 23–25]. All these aspects shape the person's perspective and behavior at any given moment.

2.2 Affordances and Signifiers

Affordances refer to the action possibilities that an object or environment offers, based on its inherent properties and the user's capabilities [15]. This now well-known concept, first introduced by Gibson [5], emphasizes that affordances are not inherent only to the object but emerge through the interaction between the object and the perceiver. This interaction-centered view is critical in understanding how objects and environments can guide human behavior.

In the context of a self-service restaurant, the affordances of physical objects—like trays, plates and food displays—inform the diner's interaction with them. A tray affords carrying multiple items, while a plate affords holding food. Importantly, the layout and design of the dining space must provide affordances that align with the diner's capabilities, such as easy access to food counters, clear paths for movement and waste disposal areas.

A signifier is a feature, signal, or cue that indicates the presence of an affordance, guiding users on how to interact with an object or interface [4, 15]. While affordances exist whether perceived or not, signifiers communicate affordances more explicitly, reducing ambiguity in user interactions [7]. They can take visual, auditory, or tactile forms, providing critical feedback to the user on what actions are possible and how to perform them. The visual cues, LED light strips, used in the modified lunch line were developed as signifiers of the affordances of the scales. They direct diners to use them correctly, to obtain nutritional information.

2.3 Cognitive Load Theory

Cognitive Load Theory (CLT) proposed by Sweller [21, 22] describes the capacity and duration limitations of the working memory when processing new information. Typically it can hold between five and

nine pieces of information at once [14], and only for approximately 18 seconds [16]. These limits are smaller when the information is complicated or interconnected. Cognitive load, the mental demands a task places on the cognitive system, encompass both task-based mental load (imposed by task or environmental demands) and user-based mental effort (the cognitive resources allocated to meet those demands). Both mental load and mental effort impact user performance [21].

2.4 Relationship of Context, Affordances and Cognitive Load Theory

The relationship between context, affordances and CLT is crucial for understanding how people interact with their environments. Each of these concepts influences how users perceive and interact with objects or interfaces, as well as the mental effort required to do so: *context* influences how these affordances are perceived and what actions are considered appropriate or possible; *affordances* provide cues about how to interact with objects or interfaces; and *CLT* explains the limits of reasonable mental effort to process information, and encourages the reduction of unnecessary complexity and load.

3 Case Study: Flavoria® Research Platform

The study is conducted in Flavoria® multidisciplinary research platform consisting of a self service restaurant with multiple lunch lines, a grill, café, and a snack shop. The dining space is an open area located in a building dedicated to medical teaching, research, and diagnostic facilities in Turku, Finland. The space features four individual lunch lines, with the main lines being double-sided for increased volume. Each main lunch line consists of eleven meal component stations: four for salads, six for side dishes and main courses and the last one for sauces.

Each meal component (salad, potatoes, chicken wings, etc.) has a dedicated scale. To connect components into a whole meal and track the customer across the restaurant on key points we utilize RFID (Radio Frequency IDentification) tags on customer trays and have readers for them on each station [11]. Above each station there is a screen that shows meal information and the state of the scale. Figure 1 shows two meal component stations, one with a tray and another with a set of keys (for comparison) on the scale.

3.1 Aiming for better accuracy with LED strips

Flavoria's measuring system captures only half of the lunch sessions correctly, 51% according to Flavoria's lunch line databases; we lose half of the diners' information because they do not use the lunch line as intended. To guide the diners to pay better attention to the scales, we installed LEDs to the first meal component station of the most popular lunch line according to data and observations detailed in Section 4. We use the mirror side of the modified lunch line as control without modifications. For additional guidance we installed an addressable LED strip with 60 LEDs/m. Additional technical details on the LED strips are provided in Appendix C. The strip is placed between the trays and food containers (Figure 1).

Figure 1 shows the four possible states and the visual cues: (1) no weight detected—no lights; (2) stable weight, no tray RFID detected, indicating either no or an incorrectly positioned tray—the light

blinks rapidly; (3) unstable weight—the lights flow smoothly from right to left; and (4) stable weight, tray RFID detected, measurement is completed and the diner can move on—the light has a very smooth subtle wavy animation. The reasoning behind the chosen visual cues was to first catch the attention of the person incorrectly using the line, then attempt to guide them to correct the misplacement. And to do these both without noticeably disturbing the diner.

4 Results and Analysis

For analysis, we collected data through observing diners in Flavoria during lunch time during six different days. Five researchers observed diners in specific locations, taking notes of what for later analysis. We also conducted two focus group interviews, both with three participants, with the aim of understanding different behaviour and decisions in regard to Flavoria lunch line, especially relating to its technology. One group consisted of first-time diners, the other of diners familiar with similar research environments (one participant had visited Flavoria before).

All participants navigated the lunch line with LED lighting without further instructions. The focus group sessions were recorded, transcribed, and participants' identities were anonymized using pseudonyms.

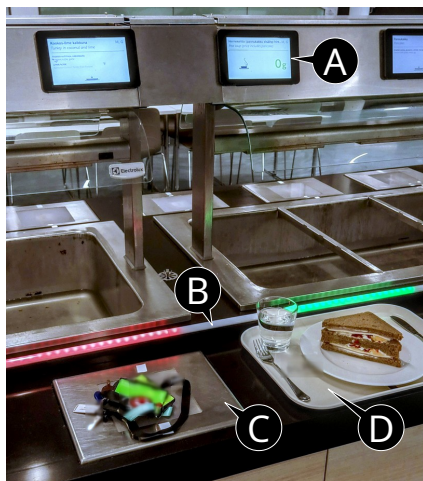
As an attempt to account for the psychological context, we had three participants complete an emotion wheel (with 114 emotions derived from six core emotions) adopted from Plutchik [18] upon their arrival. The participants demonstrated a wide range of emotions. For example, focus group diner#2 felt safe, passionate, interested, proud, brave, creative and challenged—seven emotions from two positive emotion sectors (joy, genius)—and focus group diner#3 felt grateful, safe, relief, in flow, motivated, skeptical, tired, stressed—eight emotions from five sectors (positive: joy, genius; negative: anger, sad and fear). We did not use the emotion wheel in later stages of the study; it is too laborious for studying a large number of diners in real time.

As a multidisciplinary team, we employed various tools for evaluating the guidance system we had built in a self-service restaurant, and to study diners' behavior. We engaged in frequent collaborative discussions to determine which tools were most appropriate for these different evaluation tasks.

4.1 Context Analysis

The following chapters analyze the context from the perspectives of the physical dining space, as well as the diners' subjective psychological and socio-cultural aspects. An example framework for the lunch line context is presented in Table 3.

4.1.1 Physical Context of the Dining Space. To define the key characteristics of the dining space, a thorough examination of its physical elements and observations of diner wayfinding behavior were conducted. According to that study, it has been observed that Flavoria's spatial layout naturally guides people toward the first lunch line, largely due to its prominent location near the main circulation path of the building. Additionally, the space is rich in physical stimuli, including various elements, sound levels, and kitchen noise. While the lighting is adequate, background noise from conversations and kitchen activity increases during peak hours, which can affect cognitive load [13].



State	LED View	LED Animation
(1) No Weight Detected		N/A
(2) Stable Weight, No RFID		 0.7s
(3) Unstable Weight		 1.5s
(4) Stable Weight, RFID Detected		 1.7s

Figure 1: Photo of two Flavoria research lunch line stations, showing (A) a meal component station screen, (B) an addressable LED strip in a waterproof housing with a frosted diffuser, (C) a scale platform with an RFID reader, and (D) an RFID tracked tray. The status indicators (LED strips) show an error state on the left station and success state on the right station. The table on the right enumerates the four possible status indicator states and describes how each is visualized and animated. The durations of the animations are given at the rightmost column.

As shown in Figure 2, when diners arrive to the lunch line, they first take a tray, along with necessary items like napkins, cutlery, a plate, and glasses. The food selection begins with the salad buffet, where visitors should place their trays on a scale before selecting salads. The scale ensures that the weight of the tray is zeroed out before food is added, with visitors checking the screen for confirmation. This process is repeated for each salad type, with the option to take additional items if desired. Following the salad selection, visitors proceed to choose a side dish and main dish, each requiring a similar process of placing the tray before and weighing after food is added. The system ensures that each item is accurately measured, as diners check the screens at each stage. Finally, after deciding on additional drinks, diners can proceed to the cashier to pay for their meal. Figure 2 also illustrates the items linked to the defined affordances in Section 4.2 and highlights the points where they enter and exit the journey.

4.1.2 Diner’s Psychological Context. The psychological context of a diner plays a key role in decision-making, particularly through the lens of CLT, which involves the mental effort required for tasks like choosing a meal. Cognitive load is influenced by time constraints, familiarity with the environment, the menu, and understanding of technology, as well as emotional state [17].

Time pressure can increase cognitive load, especially if diners have a limited lunch break, requiring quick decisions. Conversely, a flexible schedule allows for more thoughtful choices, reducing cognitive strain. Familiarity with the restaurant and menu also impacts cognitive load: new diners face higher cognitive demands, while regulars navigate the environment with ease. Reviewing the menu in advance reduces cognitive load, whereas encountering it at the lunch line can increase the mental effort, especially if options are unfamiliar.

Understanding the use of technology, such as scales and screens, also affects cognitive load. Unfamiliarity can cause confusion and increase mental effort, while greater awareness makes interactions smoother. Additionally, emotional state influences cognitive processing; stress can heighten cognitive load and hinder decision-making [9], whereas positive emotions can ease the process.

Overall, factors like time pressure, familiarity, technology use, and emotions shape diners’ cognitive load, impacting their dining experience and decision-making. Addressing these elements can help create a more efficient and pleasant dining environment.

4.1.3 Socio-Cultural Context. We pay attention to three aspects in the socio-cultural context: occupancy levels at the lunch line, the types of people encountered, and the level of human interaction.

Occupancy levels vary, ranging from a single diner (alone) to small groups of 2-3 people during off-peak times (10:30-11:00 and 12:30-13:30). During peak hours (11:00-12:00), the line is often crowded, with diners needing to wait for others ahead or feeling pressure from those behind.

Diners may encounter other diners (both known and unknown) or kitchen staff. Most of the lunch line is self-serve. Staff interactions occur mainly at the payment point or when staff members replenish dining tools or food containers. These brief encounters offer diners the opportunity to ask questions, though this is uncommon during the short time spent in the lunch line.

4.2 Analyzing Affordances

The analysis of affordances, of the physical space and objects located in the environmental context, reveals how diners interact with objects in the self-service restaurant and how design can improve these interactions (see Appendix B). For instance, the scale platform’s affordances include *placeability*, *slideability*, and *measurability*, allowing diners to weigh their food. However, diners

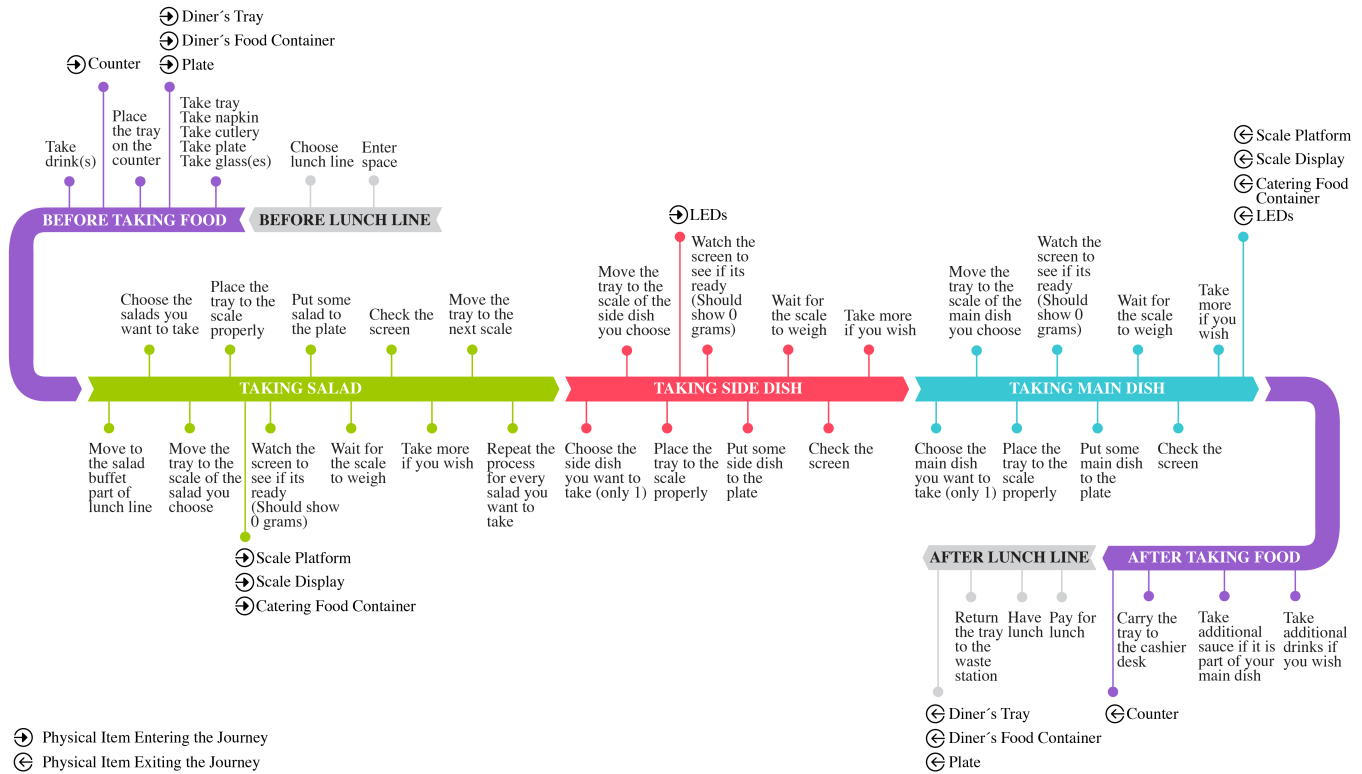


Figure 2: Lunch Line Journey

often ignore or misuse the scale, suggesting that while it has the potential for "weightability," this is not effectively communicated.

To address the ineffective communication, an LED guidance system was introduced using green lights for correct tray placement and red for incorrect. Despite using familiar traffic light colors, this system did not consistently improve user behavior. Despite the *correctability* and *alerting* affordances of the LEDs, it shows that familiar visual cues alone may not be enough when diners are multitasking or unaware of the action's importance. Alternative feedback methods, such as sound or haptic cues, may be needed.

The scale's display, meant to provide weight feedback, is often overlooked. It affords *readability* and *attention*, it competes with diners' focus on food selection. Improving its integration into the dining process, such as projecting weight data directly onto the tray, could enhance its effectiveness.

While the LEDs afford *correctability*, *processability*, and *acceptability*, they are insufficient to guide diner's behavior. This study shows that familiar signifiers like traffic light colors are not always effective in complex or distracting environments. Therefore, experimenting with alternative signifiers—such as sound, vibration, or more prominent visual cues—could be an avenue to improve user compliance and interaction.

4.3 Toolkit for Analyzing Context and Affordances in Self-Service Restaurants

Based on the findings of this study, we propose the development of a toolkit that researchers, designers, and managers of self-service dining facilities can use to better analyze and understand the context and affordances. The toolkit consists of different methodological and theoretical tools presented in Table 1. Furthermore, such a toolkit would allow them to:

First, to identify key physical objects and their affordances. A structured analysis of each object's affordances should be performed, similar to the one outlined in Table 4. This includes both primary and secondary affordances (e.g., the tray's *placeability* and *slideability*).

Second, to evaluate the effectiveness of signifiers. Signifiers should be critically evaluated to determine whether they adequately communicate the affordances. This requires observing actual user behavior to see if the intended actions are performed, and if not, identifying why the signifier failed. For example, are diners able to interpret the red and green LEDs correctly in the lunch line context? Like affordances, signifiers are outlined separately in Table 2.

Third, to consider multimodal signifiers. Where single-mode signifiers (e.g., just visual cues) fail to elicit the desired behavior, alternative signifiers should be tested. Adding auditory feedback, such as beeps when the scale is correctly used, or tactile feedback, like vibrations when the tray is misaligned, could reinforce the LED signals and make the affordances more apparent.

Table 1: Toolkit for Evaluating Guidance Systems in Self-Service Restaurants

PHASE	TOOL	DESCRIPTION
Pre-Install	Affordance Mapping	Identify the context of use by mapping the physical, social, and psychological aspects that affect diner behavior.
	User Journey Mapping	Capture what happens in the context, focusing on how diners interact with the system throughout their journey.
	Wayfinding Analysis	Analyze how easily diners can navigate and understand the guidance system within the space.
	Floorplan Validation	Assess the spatial layout to check if the physical setup supports the desired interaction.
Post-Install	Emotion Analysis	Understand diners' emotions during interactions to assess satisfaction or frustration.
	Empathy Mapping	Deepen understanding of diner behavior by exploring their needs, thoughts, and pain points.
	Data Analysis	Use metrics to validate the performance and effectiveness of the guidance system (e.g., accuracy of scale usage).
Continuous Improvement	Surveys	Gather insights into any unknown or unexpected factors that impact user experience.
	Focus Groups	Listen to diners' feedback to understand pain points and uncover areas for improvement.
	Observations	Observe users from a point where they don't know they are being observed to build understanding of user behavior

Fourth, to iterate and test. Affordances and signifiers should not be treated as fixed but rather as components of an evolving system that requires continual testing and optimization. Conducting A/B tests with different types of feedback (e.g., varying the intensity or placement of LEDs) could provide valuable insights into how to enhance user interaction.

In conclusion, analyzing affordances in self-service lunch line environment helps us understand where physical objects and digital systems succeed or fail in guiding diner behavior. This knowledge allows us to improve the design and implementation of systems like LED light guidance, ultimately creating a more intuitive and efficient dining experience.

4.4 Analysis of Data

A session is created for each diner tray as they pass through the lunch line. The session includes meal component measurements taken from each food station the diner used. A session is called synthetic if no weighing measurements were collected. Data, collected and measured since the beginning of 2024, consists of 50000 collected tray sessions for the targeted lunch lines. The new changes were presented to diners the first time on 2024-03-06 and the system has been running ever since with minimal interruptions; the longest recorded downtime, on 2024-05-06, has been three days.

In order to evaluate the effectiveness of our changes based on the session data, we calculated how many sessions out of all sessions for each lunch line included a hot meal. Based on previous data we deemed only an average sustained 60% success rate to be meaningfully improving the lunch line. This percentage would clearly rise

over the old baseline. To additionally establish a control, data was also collected on the mirror side of the modified lunch line.

Based on the results seen on Figure 3 there is no point in time exceeding 60% success rate (apart from the very end of the graph which is an artifact of only a few observation days from October). Moreover, there is no significant difference between the control and modified lunch lines. The percentage of people measured to have taken a hot meal component including all session data was 49.45% for the control lunch line and 49.53% for the modified line including all of 2024 session data and taking into account synthetic sessions.

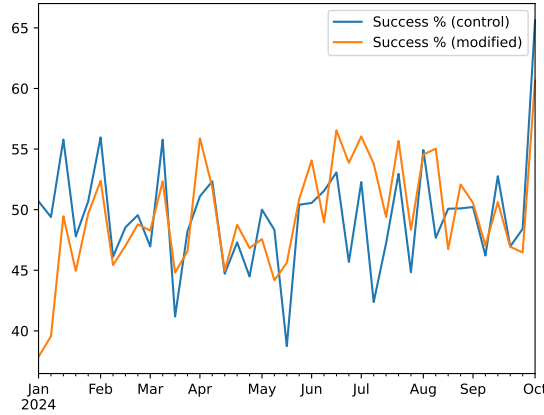
5 Discussion

The case study highlights the complexity of gathering accurate data from diners at the Flavoria restaurant. LED strips were installed to guide diners to use Flavoria's lunch line correctly, but our data revealed that the LED strips did not significantly influence the diners' behavior. The key contribution of this research is the development of a toolkit that explains why the LED strips were ineffective.

The study was conducted by a multidisciplinary team, combining expertise in architecture, service design, serious games, and software engineering. This collaboration allowed for a rich analysis, combining diverse theories and methods to help understand phenomena at Flavoria. Through shared insights, the team identified why the intervention worked or did not work. Using mixed methods, the team gathered necessary evidence for the study. This approach required open communication and a willingness to engage in deep dialogue. The resulting toolkit is adaptable, offering tools

Table 2: Signifiers Amplifying Affordances

ITEM	SIGNIFYING AFFORDANCE	TARGET
Red Blinking LEDs	Correctability, Alert, Incorrect State	Scale
Yellow Waving LEDs	Waitability, Processability	Scale
Green Pulsating LEDs	Success, Acceptability	Scale
Display	Guidance with text, multiple affordances	Mostly Scale
Yellow Waving LEDs with direction	Direct-ability (Unwanted affordance)	Queue

**Figure 3: Weekly percentage of all diner sessions with a measured hot meal component for both control and modified side of the target lunch line.**

for use before, during, and after new interventions, with potential for further expansion across different contexts.

To guide the evaluation of long-term compliance with an indicator system, an emotion mapping can be utilized. Additionally, an observer’s own internal emotional landscape affects how they recognize emotions. Individuals with clear, distinct internal emotional experiences are better at recognizing and visually representing emotions in themselves and in others, while others struggle differentiating their emotions [10]. However, based on our attempt to emotional evaluation at the focus groups, it was evident that there are a wide range of possibly relevant emotions and it might be infeasible to account for their impact. Thus, we decided to shift from trying to understand the emotions and their sources into cognitive load the experienced emotions bring upon a diner at the lunch line.

Additionally we suggest empathy as a toolkit tool to understand the diners. Framework for Empathy in Design suggests that empathy consist of two elements, affective component which includes emotional response and the cognitive component which includes understanding the meaning [12].

As a multidisciplinary team, we encountered confusion over terminology. A similar study should regularly clarify terms during the research process and ensure all researchers are familiar with the evolving vocabulary.

A multidisciplinary approach also allowed us to identify areas of analysis that require further consideration. For instance, more

in-depth reflection is needed to determine the appropriate tools for studying human behavior influenced solely by an object’s affordance. In the context of SSR, for example, behavior might also be shaped by the presence of other diners, especially during busy hours. These dynamics are complex and challenging to distinguish through observation alone.

In focus group observations, attention to the screens and scales for the first-time diners was observed to be scattered. One of the diners was aware of the screens and the actual weight of the food taken. This affected also how much food was taken, which revealed that there were challenges of interpretation of the scale information shown. “And then I started thinking that is it always green [the number of grams in the screen]? And also, I was thinking, am I taking too much or am I taking too little? And I kind of wanted to test if I take more, does it go to red?” (*Focus Group A diner#1*)

Moreover, Focus Group B, conducted during peak hour at Flavoria, revealed that cognitive load and the surrounding context significantly impacted the participants’ ability to interpret visual cues. Despite being aware of the technology and having experience in a similar research environment, none of the diners noticed the additional guidance. When asked about the LEDs, one participant responded, “Which LEDs?” (*Focus Group B diner#1*). Additionally, only one participant, who was not a first-time diner at Flavoria, mentioned using the scales, explaining that he wanted to check if they were functioning. The other two participants expressed that the environment was too hectic, preventing them from paying attention to the scales or other visual indicators.

6 Conclusion

This research highlights four critical takeaways from our attempt to improve the use of scales in a self-service restaurant through LED lights guidance systems. First, interventions designed to influence behavior without increasing cognitive load, like LED indicators, did not produce the expected results. While screens were avoided to reduce cognitive overload, we found that alternative methods also require careful consideration of user behavior and attention. Second, our assumption that users would interpret the LED signals similarly to traffic lights was flawed. Although the context of the traffic lights is familiar, we neglected to account for scenarios where people disregard traffic signals, such as in high-pressure or hurry situations, which mirrors the behavior we observed in the dining environment. Third, the bias introduced through limited observational data and low statistical significance led to an overestimation of the effectiveness of our solution. Small sample sizes in observations and anecdotal evidence from diner interactions skewed our understanding, resulting in a false sense of success. Fourth, utilizing a mixed-methods approach and focus group discussions, we

were able to identify limitations of using LEDs as a visual guidance system for the scales.

The toolkit we have developed addresses the above challenges by encouraging designers to rethink traditional theories and frameworks. It incorporates context analysis, affordance theory, and cognitive load theory but also improvises the pitfalls often encountered in real world applications of these theories and frameworks. This toolkit can be used to either retrospectively and objectively analyze success/failure or proactively guide the design of new systems leading to future interventions that can be better optimized for real-world success, ultimately enhancing user experience and system effectiveness.

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A Lunch Line System Context

An example framework for the lunch line context is presented in Table 3.

B Self-service Lunch Line Affordances

We use affordances theories to map the new system in Table 4.

C Software and Hardware Description of LEDs

Addressable WS2815/SMD5050 LEDs are used for their low voltage drop and size. The low voltage drop eliminates potential brightness differences towards the end of the LED strip. The strip is housed in a waterproof smooth case with a frosted D-shaped flexible diffuser ontop and then further doubled-enclosed in a IP67 waterproof rating clear silicone (inside the diffuser) to prevent any water ingress. A 12V 120W waterproof power supply is used for the roughly 4-5 meters of installed LEDs to allow driving the LEDs at full brightness for maximal visibility.

The LEDs are connected to a WT32-ETH01 ESP32 microcontroller and networked via ethernet to eliminate WiFi interference. WLED firmware is used for animations and for addressing the LEDs. The meal component stations use MQTT to communicate with a central python service where the station states are encoded into WLED states and further transmitted again via MQTT to WLED. The setup has proved reasonably robust even under load with no observed latency fluctuations for the studied 6 scales. Additionally, the system can be scaled up without limits by adding multiple microcontrollers. WLED configuration files are provided on request.

For hygiene reasons the strips are presently limited to the main course stations: Salad stations are not included as this would require running cables throughout the whole lunch line preventing easy cleaning. For a comprehensive installation, cable holes would be drilled through the lunch line for each separate strip part and connected electrically under the lunch line.

Table 3: Framework for Lunch Line System Context: Main Context Categories, Themes, and Scenarios

CATEGORY	CONTEXT THEME	CONTEXT SCENARIO
Environmental & physical (dining space)	Physical objects	Lunch line counter, diner's trays with RFID tags, plates, cutlery, napkins, glasses, food in metal catering containers, scale platforms, scale displays showing meal details and allergens, serving ladles, handwritten labels
	Environmental factors	Lights, noise, temperature
	Weekday	Mon, Tue, Wed, Thu, Fri
	Timing of lunch	Before peak hour 10:30–11:00, Peak hour 11:00–12:00, After peak hour 12:00–13:30
Psychological (diner)	Time available for lunch	Limited (e.g., fixed time), Flexible (timing and length)
	Familiarity of the restaurant	First-time diner, Occasional diner, Regular diner
	Familiarity of the menu of the day	Viewed prior to arrival, Viewed at the beginning of the lunch line, Physically observed the lunch line
	Awareness of the scales and screens	Unaware, Partially aware, Aware but ignorant, Aware and compliant
	Awareness of the LEDs	Unaware, Partially aware, Aware but ignorant, Aware and compliant
	Emotional status upon arrival	Emotional wheel tested to reveal emotions <i>Note: Challenges arise from only observing, without knowing why and what causes the emotions.</i>
Socio-cultural	Occupancy at the lunch line	Alone, 2-3 diners: Someone behind but not ahead (pressure to move forward), 2-3 diners: Someone ahead but not behind (waiting for others to proceed), Crowded (people queuing behind and ahead)
	Social context	Other known diners (e.g., friends, colleagues), Other unknown diners, Kitchen staff members
	Human-to-human interaction	No interaction, Non-verbal interaction (e.g., eye contact, nodding, smiling), Verbal interaction (e.g., greeting, talking), Both verbal and non-verbal interaction

Table 4: Affordances of the self-service lunch line from the diner's point of view

KEY PHYSICAL OBJECTS	AFFORDANCES
Scale platform	Placeability
	Slideability for correcting the position
	Liftability of the objects above the scale
	Area of transparency on the scales
	Weightability
	Measurability
Scale display	Readability
	Attention attractability
LEDs	Attention attractability
Catering Food Container	Takability
Plate	Slidability
	Containability
Counter	Horizontal slidability
Diner's food container (lunch box or tray)	Holdability
	Liftability
	Containability
	Slidability
Diner's tray	Traceability